Bellabeat Case Study: How Can a Wellness Company Play It Smart?

Insights from Fitbit Fitness Tracker Data

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Role: Data Analyst

Tools: Excel

The Goal: Unlocking Growth for Bellabeat

Analyze smart device usage data to identify trends in user activity behavior
and provide actionable insights to support Bellabeat’s marketing
and product strategy.

Key Questions

- How active are Bellabeat users on a daily basis?
- Do activity patterns differ between weekdays and weekends?
- How does activity intensity relate to calories burned?
- What behaviors can Bellabeat encourage to improve user engagement?

Data Overview

Data Source: Fitbit Fitness Tracker Dataset (Google Data Analytics Capstone)

Time Period: March–April 2016

Key Tables Used

- Daily Activity
- Activity Intensity
- Calories
- Steps
- Sedentary Minutes

Tools

Microsoft Excel (data cleaning, pivot tables, charts)

Data Cleaning & Processing

Steps Taken

- Standardized date formats
- Removed duplicate records
- Validated inconsistent entries
- Flagged abnormal records (e.g., zero steps with activity minutes)

Result

Clean, consistent dataset ready for analysis

Key Insights

Insights 1

Key Metrics (Daily Averages)

- Average Steps: **6,561**
- Average Calories Burned: **2,184 kcal**
- Average Sedentary Time: **996 minutes**

Observation: Most users are moderately active but highly sedentary, falling below commonly recommended daily step targets.

Weekdays vs Weekends

Insight 2

DaysType	Average of TotalSteps	Average of Calories	Average of SedentaryMinutes
weekdays	6550	2165	996
weekends	6584	2224	996
Grand Total	6561	2184	996

Observation: User activity patterns are almost identical on weekdays and weekends.

Activity Intensity

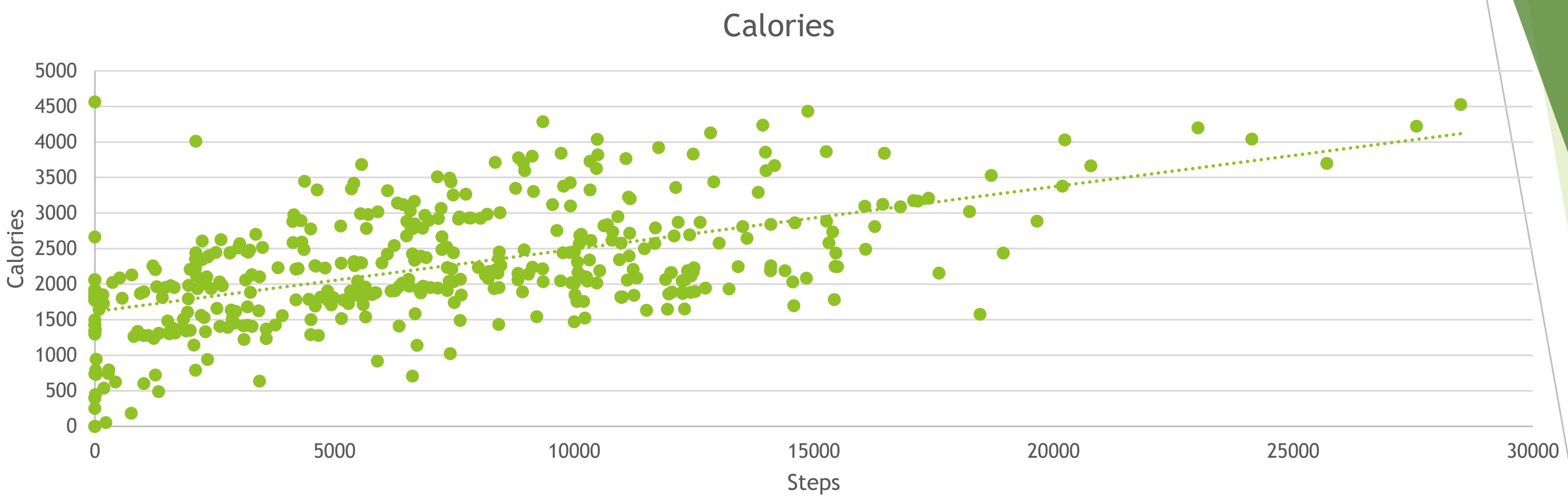
Insight 3:

Average Minutes per Day

- Sedentary: 996
- Lightly Active: 170
- Very Active: 17
- Fairly Active: 12

Most user time is spent sedentary or lightly active.
High-intensity activity is very limited.

Steps vs Calories



Observation

Higher steps generally lead to higher calories burned but relationship is not linear

Key Finding

Some users burn more calories with fewer steps, indicating that: Activity intensity, Speed, Non-step activities play a significant role.

Conclusion

Calories burned depend more on intensity than step count alone.

Dashboard


6561
Total Steps
996 (mins)
Sedementary Minutes
2184 (Kcal)
Calories Burned
199 (mins)
Active Minutes
Steps Vs Calories
5000
4000
3000
2000
1000
0
0
5000
10000
15000
20000
25000
30000
Behavioral Distribution
1200
1000
800
600
400
200
0
Total
Average of VeryActiveMinutes
Average of FairlyActiveMinutes
Average of LightlyActiveMinutes
Average of SedentaryMinutes
Do User behavior varies on weekdays to weekends?
7000
6000
5000
4000
3000
2000
1000
0
weekdays
weekends
Average of TotalSteps
Average of Calories
Average of SedentaryMinutes
Day_Type

Analysis Summary

- Users are moderately active but highly sedentary
- Weekday and weekend behaviours are nearly the same
- Step count alone does not explain calorie burn
- Activity intensity is a stronger driver of energy expenditure

Recommendations

1. Shift Focus from Steps to Intensity
Encourage short bursts of moderate-to-high intensity activity, not just step goals (e.g short workout)

2. Reduce Sedentary Time
Introduce reminders to break long inactive periods.

3. Promote Habit-Based Engagement
Since activity patterns are consistent across days, focus on daily routines rather than weekend challenges.

4. Highlight Intensity Metrics in the App
Display active minutes and intensity levels alongside step counts.

Conclusion

This analysis shows that Improving user health outcomes requires focusing on how users move, not just how much they move. By emphasizing intensity and reducing sedentary behavior, Bellabeat can drive more meaningful engagement and value for users.